FLA Project – Java Implementation

Develop Java programs to implement the following:

1. DFA Simulation

Simulate a given DFA and determine whether a given input string is accepted or rejected.

2. NFA to DFA Conversion

Convert a given NFA into an equivalent DFA using the subset construction method.

3. DFA State Reduction

Minimize a given DFA by identifying and merging equivalent states.

4. Intersection of Two DFAs

Given two DFAs, construct a DFA that accepts the intersection of their languages:
$L = L_1 \cap L_2$
using the product construction method.

5. PDA Simulation

Simulate a given Pushdown Automaton, including its stack operations, and determine whether a given input string is accepted or rejected.

All five tasks must be implemented in Java.